Supplementary Material 7: Tuning to decide on best binwidth for viewing in p-D

The model fitting depends on several parameters, including the hexagon bin parameters and the low-count bin removal process. The hexagon bin parameters define the bottom-left bin position (s_1, s_2), the number of bins in the horizontal direction (b_1), which determines the number of bins in the vertical direction (b_2), the total number of bins (b), and the number of non-empty bins (m). Low-count bins are removed using standardized bin counts, defined as $w_h = n_h/n$, for $h = 1, \dots, m$.

Default values are provided for these parameters. In particular, the default number of bins is determined from the sample size by setting $b_1 = n^{1/3}$, consistent with the Diaconis–Freedman rule [@freedman1981], with b_2 computed to preserve the aspect ratio of the layout.

Rather than selecting a single set of parameters a priori, we recommend a structured procedure to examine the stability of the fitted model across a range of values. A practical workflow is as follows:

1. **Select a grid of binwidths:** Choose a sequence of values for b_1 ranging from coarse to fine resolution (e.g., from small values up to $b_1 = \sqrt{n/r_2}$). This controls the level of detail in the fitted wireframe.

2. **Compute HBE across the grid:** For each choice of b_1, compute the HBE. Stable regions where HBE changes slowly indicate parameter values where the fitted model is not overly sensitive to binning resolution.

3. **Assess bin occupancy:** Examine summary quantities such as average bin count $(\bar{n})$, average standardized bin count $(\bar{w})$, and the proportion of non-empty bins (m/b). Reasonable parameter choices typically correspond to:

   - sufficiently large $\bar{n}$ to avoid sparse bins, and
   - moderate values of m/b, indicating that bins cover the data without excessive fragmentation.

4. **Check robustness to bin removal:** Evaluate the effect of removing low-count bins using a range of thresholds. Parameter values where HBE remains stable under small amounts of bin removal indicate robustness of the fitted structure.

5. **Compare across layouts (if applicable):** When comparing multiple NLDR layouts, examine HBE as a function of average bin count or standardized bin count. This helps account for differences in point density between layouts.

This process does not yield a single "optimal" parameter choice, but instead identifies a range of stable and interpretable settings. The goal is to avoid parameter values that produce unstable fits, overly sparse binning, or misleading structure in the fitted wireframe.

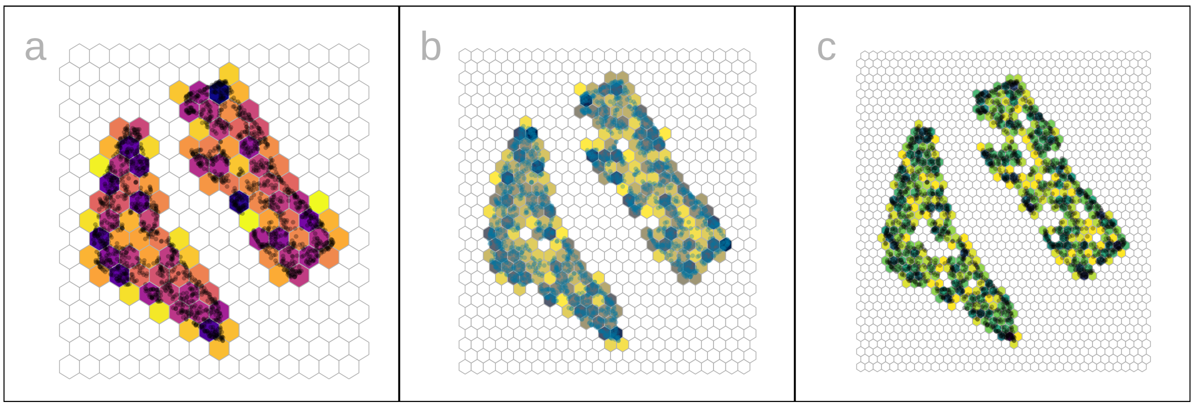

Figure 1: Hexbin density plots of tSNE layout of the 2NC7 data, using three different b_1 specifications yielding different b_2, b, m: (a) **15**, 18, 270, 98, (b) **24**, 29, 696, 209, and (c) **35**, 42, 1470, 386. Color indicates standardized counts, dark indicating high count, and light indicating low count. At the smallest binwidth, the data structure is discontinuous, suggesting that there are too many bins.

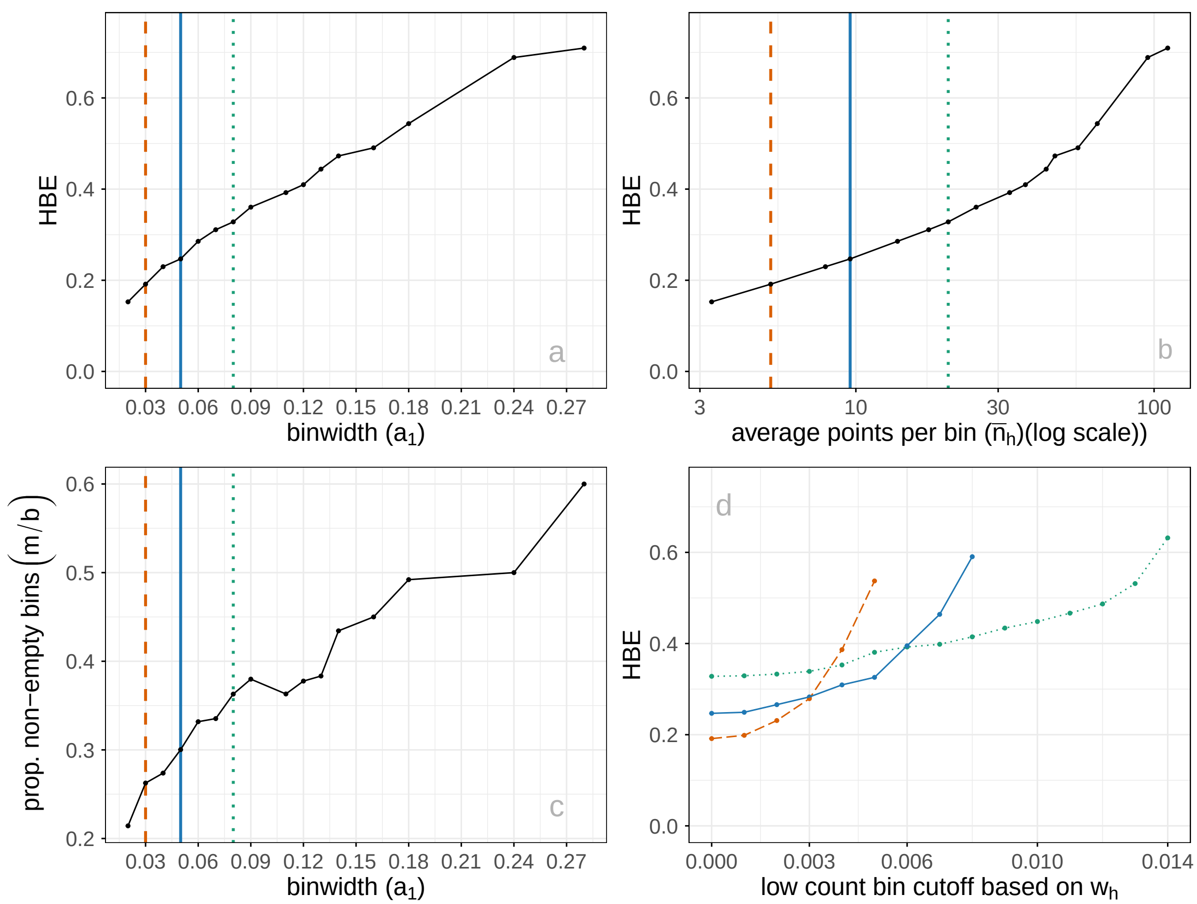

Figure 2: Various plots to help tune the model fit: (a) HBE vs a_1, (b) HBE vs $\bar{n}_h$, (c) proportion of non-empty bins (m/b) vs a_1, (d) HBE vs w_h cutoff used for removing low count bins. Color indicates the three binwidths show in Figure 6: 0.03 (orange dashed), 0.05 (blue solid), and 0.08 (green dotted). The better model fit will have low HBE, but reasonably sized bins that capture the data sufficiently. Proportion of non-empty bins tends to increase with a_1 (c). Removing a few low-count bins doesn't substantially change HBE, for all three binwidths (d).